

3.1 The characteristics of the Antarctic ecosystem

This chapter will explain:

- ★ the location of the Antarctic ecosystem
- ★ the climate of Antarctica
- ★ the features of the Antarctic ecosystem.

The location of the Antarctic ecosystem

The Antarctic refers to everything south of the Antarctic Circle ([Figure 3.1](#)) including land, ice shelves and sea, while Antarctica refers to the landmass that is the southern polar continent. Antarctica consists of two parts, East and West Antarctica. The landmass is about 14 million km², which is about 6 million km² larger than the USA. The Antarctic ice sheet covers about 98 per cent of the continent and is on average about 1.6 km thick. Antarctica contains around 90 per cent of the world's ice and 70 per cent of its fresh water. If all the ice in Antarctica were to melt, global sea levels could rise by up to 60 m.



▲ **Figure 3.1** Antarctica and the Antarctic Circle

A large part of the West Antarctica Ice Sheet (WAIS) lies below sea level and its edges are floating ice shelves. The WAIS accounts for about 10 per cent of the total volume of the Antarctic ice sheet, about 2 million km³. If WAIS were to melt, sea levels could rise 3–4 m. The East Antarctica Ice Sheet (EAIS) rests on the Antarctica landmass; it's a thicker, larger ice sheet, and is nearly 5 km deep.

In 2024 it was reported that scientists had identified a tipping point for the loss of ice sheets in Antarctica that could cause a greater rise in sea level than previously expected. They examined how warming gets between coastal ice sheets and the ground they rest on. The warm water melts some of the ice, allowing more water to flow in. This reinforces the warming. Scientists had warned in 2022 that the Greenland ice cap was in danger of collapsing, and in 2023 they found that accelerated melting in western Antarctica was inevitable for the rest of the century.

Interesting note

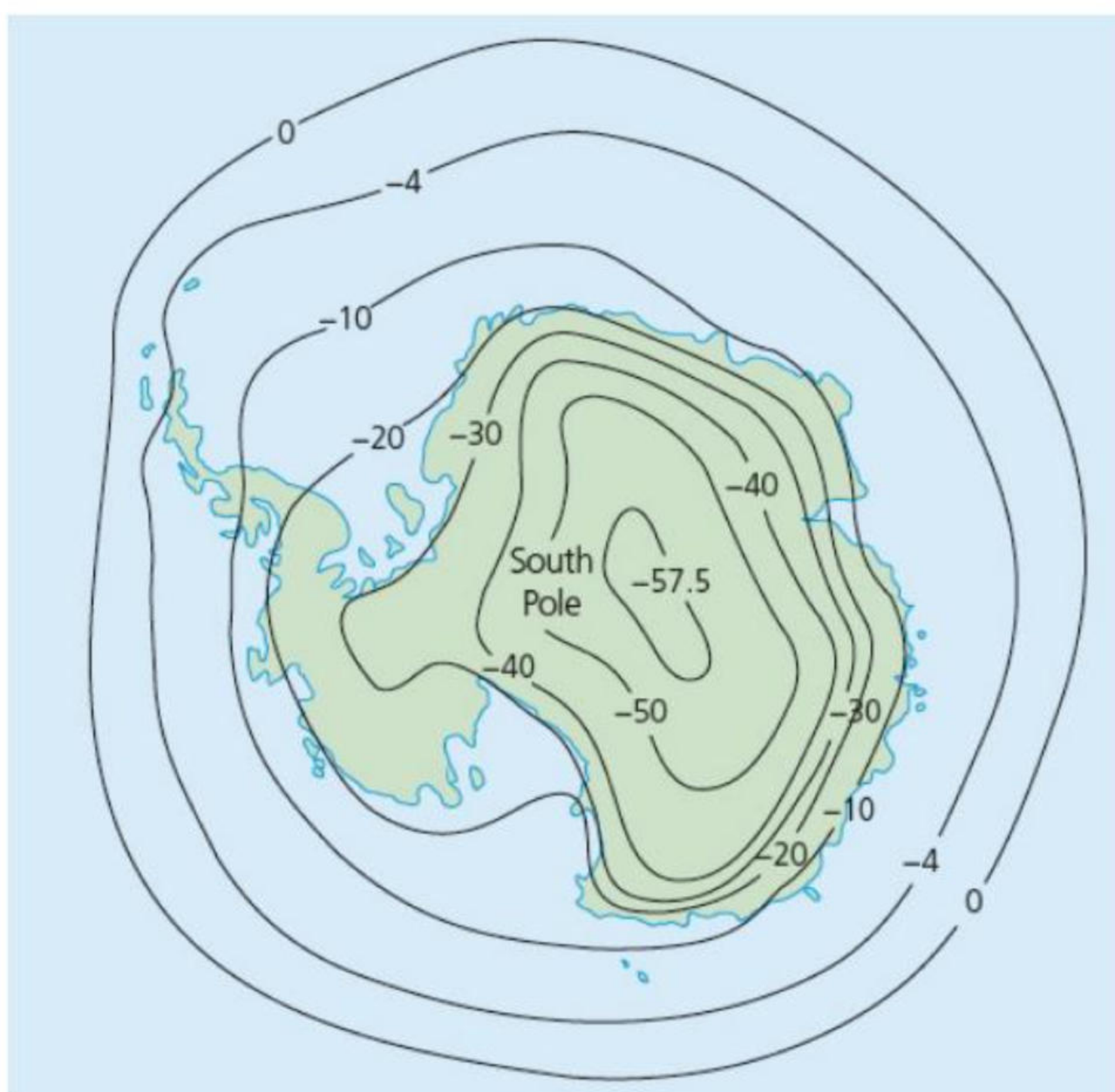
Antarctica is unique among the continents in that it is the only one with no indigenous population. The first 'indigenous' child to be born there was an Argentinian boy, Emilio Marcos Palma, born in 1978. Since then, more than ten children have been born in Antarctica. It is inhabited by around 5000 people in summer (scientists and support staff) and about 1000 people in winter.

The climate of Antarctica

Temperature

The main characteristic of Antarctica's climate, in terms of both duration and intensity, is that it is cold. On ice sheets in the interior, the winters are severe and the summers non-existent. However, in coastal areas, the winters are warmer and the summers cooler. There are many reasons why Antarctica is so cold:

- » Due to high **air pressure**, the continent is relatively cloud-free and receives little **rainfall**. High pressure areas have descending air so there is limited uplift or **cloud** formation. This allows much of the outgoing long-wave radiation to escape from the Earth's surface, making the region very cold.
- » Due to its high latitude, the low angle of the sun in relation to the ground means it receives less radiation. In the brief summer, however, there is sufficient solar radiation to melt snow and ice and to support life.
- » Due to its colour, Antarctica reflects much solar radiation. This reflectivity is called **albedo**. Dark surfaces absorb more solar radiation (up to 90 per cent), whereas fresh snow may absorb as little as 10 per cent and reflect the other 90 per cent.
- » There is a lack of dust and water vapour in the atmosphere. Dust and water vapour are important for trapping long-wave radiation from the Earth's surface, which heats the Earth's atmosphere. Hence, limited amounts of long-wave radiation are trapped over Antarctica to heat up the atmosphere.



▲ **Figure 3.2** Mean annual surface temperatures in Antarctica (°C)

Antarctica's temperatures are low throughout the year ([Figure 3.2](#)). The mean annual temperature is below freezing everywhere south of 60°S. On the continent, the mean annual temperature varies from about -60°C inland to around -10°C by the coast. There are seasonal contrasts too. On the continental interiors, temperature ranges from about -70°C in July to -25°C in January. In contrast, along the coast it ranges from -40°C in winter to -2°C in summer. The height of much of Antarctica also leads to lower temperatures. In general, temperatures fall 1°C per 100 m. Around 50 per cent of Antarctica is over 2000 m and 25 per cent over 3000 m. If those areas were at sea level, Antarctica would be 20–30°C warmer!

Winds are particularly important in Antarctica and can intensify the chilling effects of low temperatures ([Table 3.1](#)).

▼ **Table 3.1** The effect of wind speed on temperature – the wind chill effect

Air temperature (°C)	Wind speed (metres per second)				
	Calm	2.5	5	10	20
0	19	0	-7	-12	-18
-5	16	-6	-13	-19	-26
-10	13.5	-11	-19	-26	-33

-20	9	-21	-31	-40	-44
-30	4.5	-31	-43	-54	-63
-40	0	-41	-55	-68	-78

Blizzards are also a feature of Antarctica. When surface winds reach 10 m/s^{-1} , snow is picked up and saltated (bounced) along the ground. When winds reach a speed of 15 m/s^{-1} , they can pick up enough snow to reduce surface visibility to zero.

Precipitation

Precipitation levels in Antarctica are low and consist mainly of snow. Many of the central parts of East Antarctica receive less than 50 mm of water equivalent (that is, how much water the snow would produce if it fell as rain), whereas coastal areas can receive 200–600 mm. Once more, **altitude** has an impact. As moist air is forced to rise, there is condensation, cloud formation and precipitation, and maximum snowfall occurs at about 1600 m.

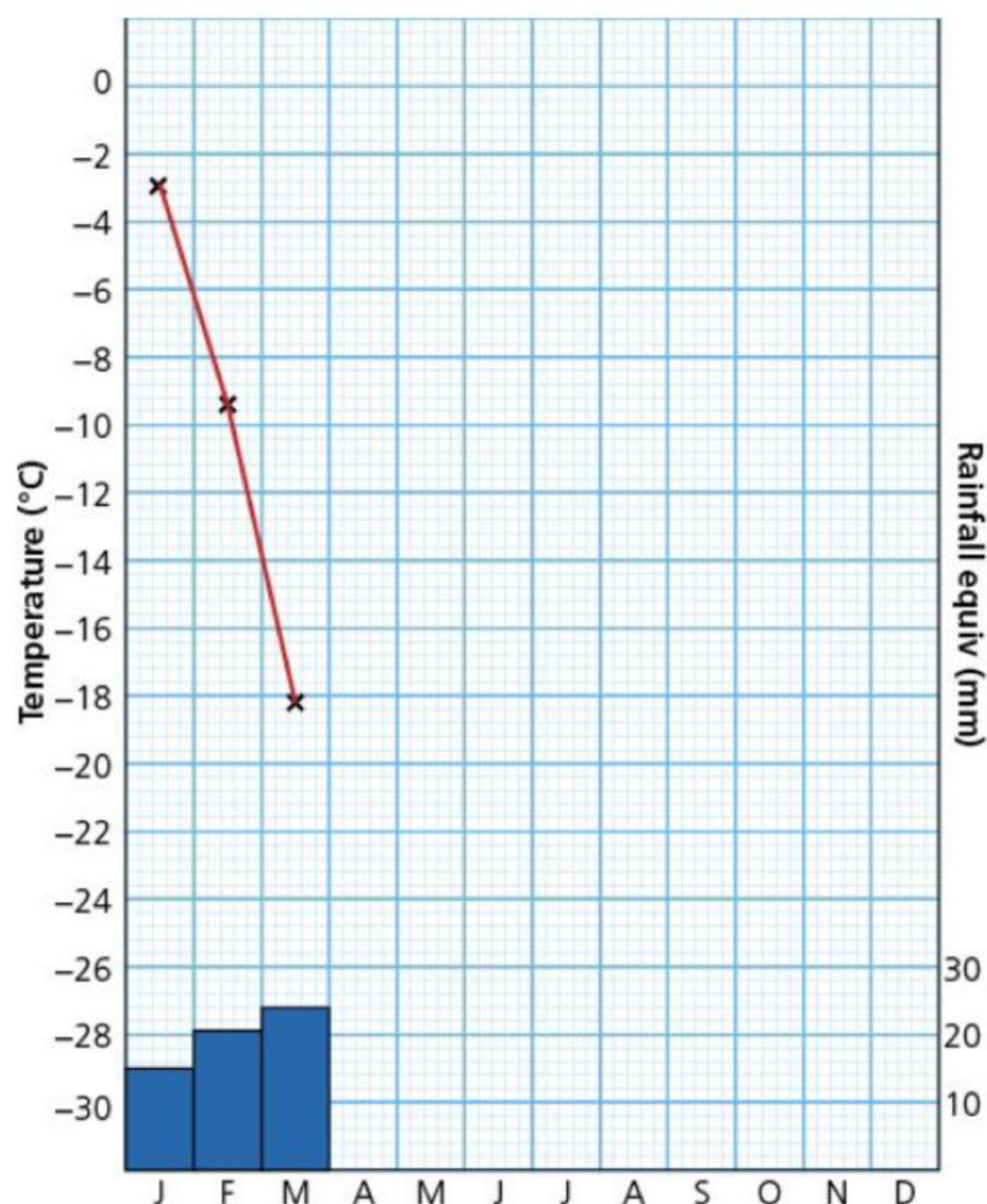
Activities

- Using [Table 3.1](#), state the wind chill when:
 - air temperature is 0°C and wind speed is 5 m/s^{-1}
 - air temperature is -10°C and wind speed is 10 m/s^{-1} .
- [Table 3.2](#) shows climatic data for McMurdo, Antarctica.
 - Calculate the mean annual temperature and the total annual rainfall.
 - Explain why higher temperatures occur between November and February.

▼ **Table 3.2** Climatic data for McMurdo, Antarctica

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Annual
Temperature ($^{\circ}\text{C}$)	-2.9	-9.5	-18.2	-20.7	-21.7	-23.0	-25.7	-26.1	-24.1	-18.9	-9.7	-3.4	
Rainfall equivalent (mm)	15.0	21.2	24.1	18.4	23.7	24.9	15.6	11.3	11.8	9.7	9.5	15.7	

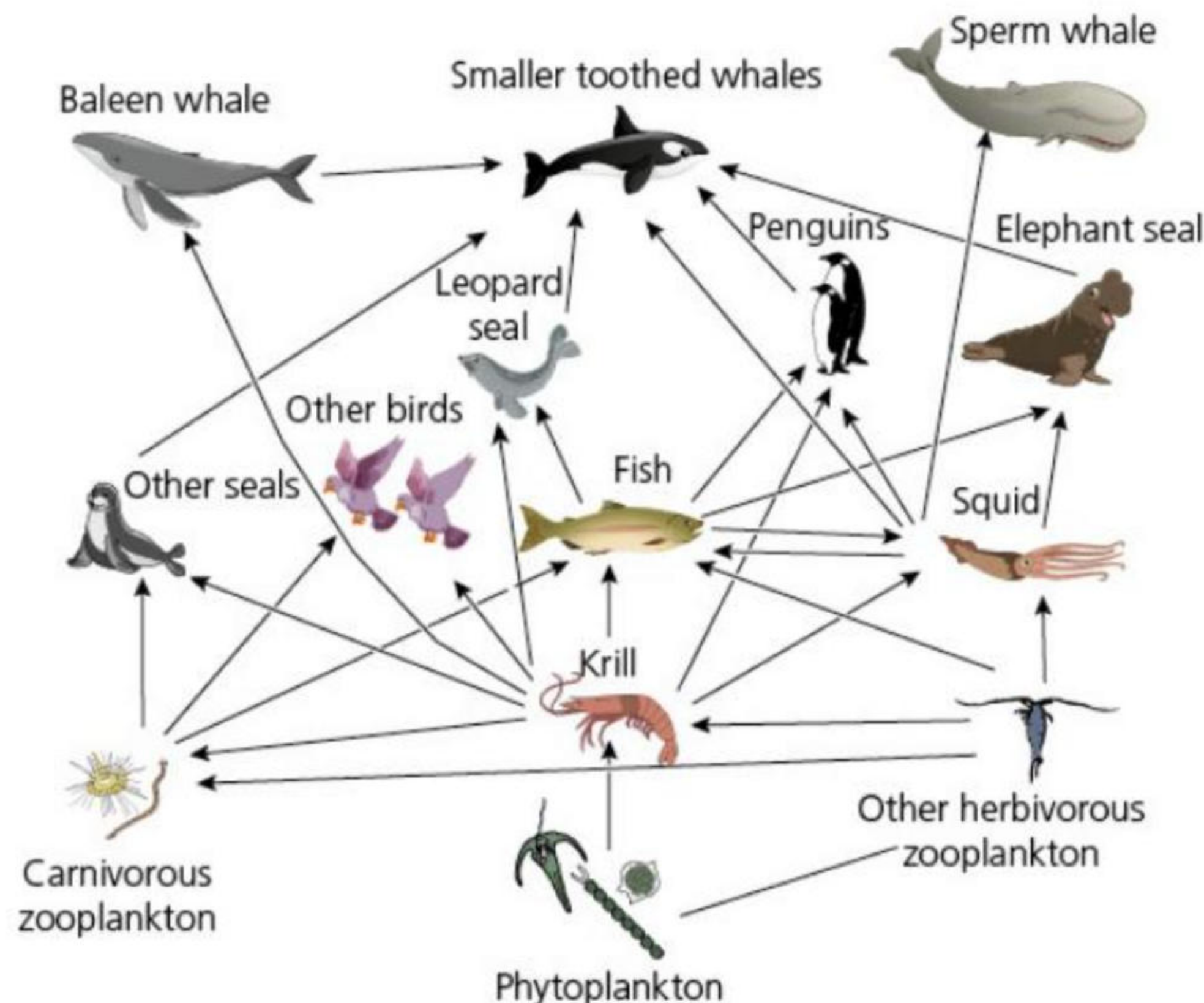
- Copy and complete the climate graph for McMurdo, Antarctica shown in [Figure 3.3](#).



▲ **Figure 3.3** Partially completed climate graph for McMurdo, Antarctica

The features of the Antarctic ecosystem

Despite the low temperatures, limited moisture, poor **soils** and lack of sunlight for six months, there are many **ecosystems** in and around Antarctica ([Figure 3.4](#)). There are around 200 species of lichen, more than 100 species of mosses and liverworts, 30 species of macrofungi, and large numbers of algae. Although the growing season is short, many species exist in the milder parts of the Antarctica Peninsula. Opportunities for photosynthesis and water availability are extremely limited, and growth rates and reproduction are very low, and some species, notably glass sponges in Terra Nova Bay on the Ross Sea, can live for up to 15,000 years.



▲ **Figure 3.4** Ecosystems around Antarctica

Antarctica's marine and aerial environments support an abundance of species, including penguins, seabirds, seals, whales and krill. The Southern Ocean contains some of the coldest sea water in the world. Cold water is very dense and sinks, displacing lower water to the surface. This upwelling water brings up oxygen and nutrients to the surface. This provides nutrients for surface algae, which then provide food for krill, a shrimp-like marine crustacean, which in turn provide food for fish, seals, whales and sea birds. Marine ecosystems in Antarctic waters may be relatively simple but they are, nonetheless, varied.

Krill ([Figure 3.5](#)) are about 5–6 cm long and feed on phytoplankton and zooplankton. Their biomass in Antarctic waters is estimated at around 500 million tonnes. Krill is commercially important, and between 150,000 and 200,000 krill are harvested in the Southern Ocean each year.



▲ **Figure 3.5** Krill

The Southern Ocean has over 250 species of fish that can survive temperatures below freezing, such as the Antarctic icefish. The Patagonian toothfish declined by about 50 per cent between 1985 and 2020 due to overfishing. Many species of whale are seasonal visitors to the Southern Ocean and these were hunted intensively in the early parts of the twentieth century. Species such as blue, humpback, minke, killer, southern right and sperm whales feed and breed around the Antarctic. Commercial harvesting of killer whales (orcas) last occurred in 1979–80. The population of orcas may now be in excess of 80,000. Likewise,

seals are found in large numbers in the Southern Ocean despite having previously been harvested. Seal species present include leopard, Ross, southern elephant and Weddell. Leopard seals can be over 3 m in length and weigh up to 500 kg. They feed on fish, penguins and other seals' pups. The Southern Ocean is home to nearly 90 per cent of benthic (deep sea) fauna.

Bird life includes terns, petrels, skuas, albatrosses and flightless penguins. Species of penguin include emperor, chinstrap, Adélie and royal. Emperor penguins are the largest – an adult male can be over 1 m tall and weigh around 40 kg. The emperor penguin is the only one to breed on the Antarctic Peninsula in winter. Albatrosses can have a wingspan of over 3.5 m. There are believed to be around 20,000 breeding pairs of albatrosses around the islands of the Southern Ocean.

Activities

- 1 Study [Figure 3.4](#). Create a food chain with seven levels in it.
 - 2 Suggest why food chains in the Antarctic oceans are more developed than food chains in the Antarctic continent.
-

The interrelationships between abiotic and biotic factors in the Antarctic ecosystem, and how flora and fauna adapt to survive

As elsewhere, abiotic factors in Antarctica include climate (for example temperature, precipitation and wind), nutrients, sunlight, soils, rocks and permafrost. Biotic factors include the mammals, birds, fish, insects, mosses and lichens found in the oceans, on land and in the soil.

There is a close relationship between abiotic factors and biotic factors. The lack of sunlight for six months, low temperatures, limited soil development and high wind speeds make life in central Antarctica difficult. However, around coastal areas, temperatures may be higher and conditions less extreme, allowing more species to survive. Many species use Antarctica for specific purposes, such as egg laying and raising their young.

In contrast, the oceans around Antarctica have a plentiful supply of nutrients at the surface due to the upwelling ocean currents. These support a large amount of plankton and krill, which in turn support large and diverse food webs that include fish, seals, penguins and whales.

Even sea ice can support ecosystems based on single-cell species such as bacteria and phytoplankton. These then support grazers such as zooplankton and krill, which in turn can support fish and other carnivores.

Life on land in Antarctica is somewhat limited. Most of the plant life includes mosses, lichen and liverworts, which support about 60 microspecies of invertebrates such as springtails and nematodes. The largest permanent dweller on Antarctica is the Antarctic midge (2–6 mm long).

There are many species adaptations to living in and around Antarctica. Seals, whales and penguins have thick layers of blubber (insulating fatty tissue). Blubber contains red blood cells that help regulate blood flow to the animals' skin surface. During warm conditions, the blood vessels expand and blood cells are taken to the surface. However, in cold conditions, blood vessels contract, taking blood cells away from the surface to vital organs.

Penguins have different types of feathers. On the surface there are coarse, abrasion-resistant, waterproof feathers. Over the course of a year these get damaged, so penguins replace these feathers each summer during a 'catastrophic moult'. Other feathers include soft, downy feathers, which provide insulation near the penguin's skin surface.

Penguins also form huddles. Emperor penguins may form large-scale huddles during major storms. The temperatures in the centre of the huddle can reach 37°C. Individuals cool down by moving towards the edge of the huddle and warm up by moving towards the centre.

Antarctic fur seals have two types of fur:

- » Coarse, long-guard hairs, which protect the under-fur from water and abrasion.
- » Dense (up to 50,000 hairs per cm²) water-resistant insulating fur.

Antarctic seals' eyes are adapted for hunting in dim conditions. They have large pupils that dilate to let more light in. Ross seals' eyes can be up to 7 cm in diameter. Seals also have very sensitive whiskers, which can detect the movement of prey at large distances.

Antarctic icefish have developed anti-freeze proteins in their blood. They are the only adult vertebrate with no red blood cells. The proteins bind to ice crystals, reducing their growth and preventing them from joining other ice crystals that could damage the icefish with their jagged edges.

Some species of sea spiders, worms and crustaceans reach very large sizes. It is believed that the high availability of oxygen in cold polar waters allows some species to keep on growing. This is known as gigantism.

Activities

- 1 Describe the relationship between abiotic factors and biotic factors in Antarctica.
 - 2 Explain how Antarctic fur seals are adapted to life in the Southern Ocean.
-